Lab 7: Fuzzing for Input Validation Bugs

Software Engineering Lab | Repo: L6_L7_42_salon_booking (Lab 7)
Tool: Hypothesis (property-based fuzzing) with pytest

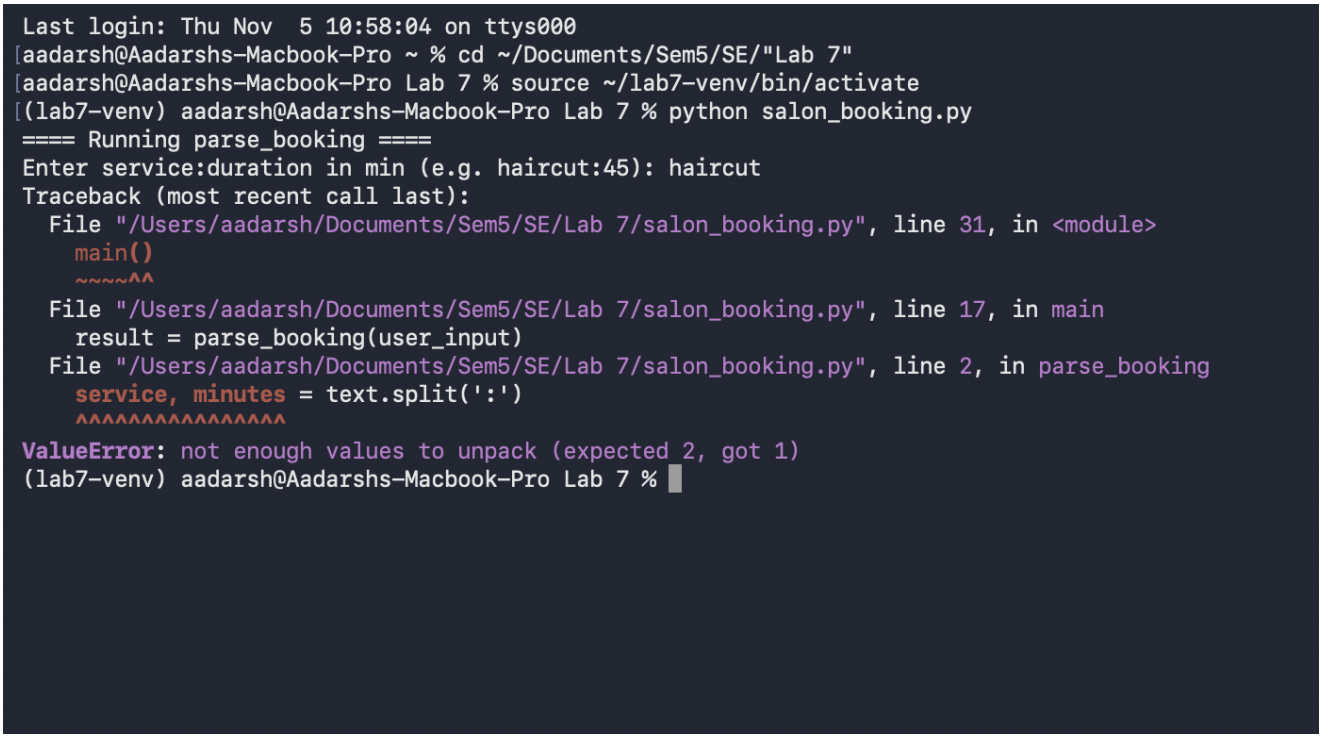
Step 1: Install dependencies

hypothesis and pytest were installed into a Python virtual environment (macOS Homebrew Python blocks system-wide pip installs):

```
python3 -m venv ~/lab7-venv
source ~/lab7-venv/bin/activate
pip install hypothesis pytest
```

Step 2: Run the buggy application

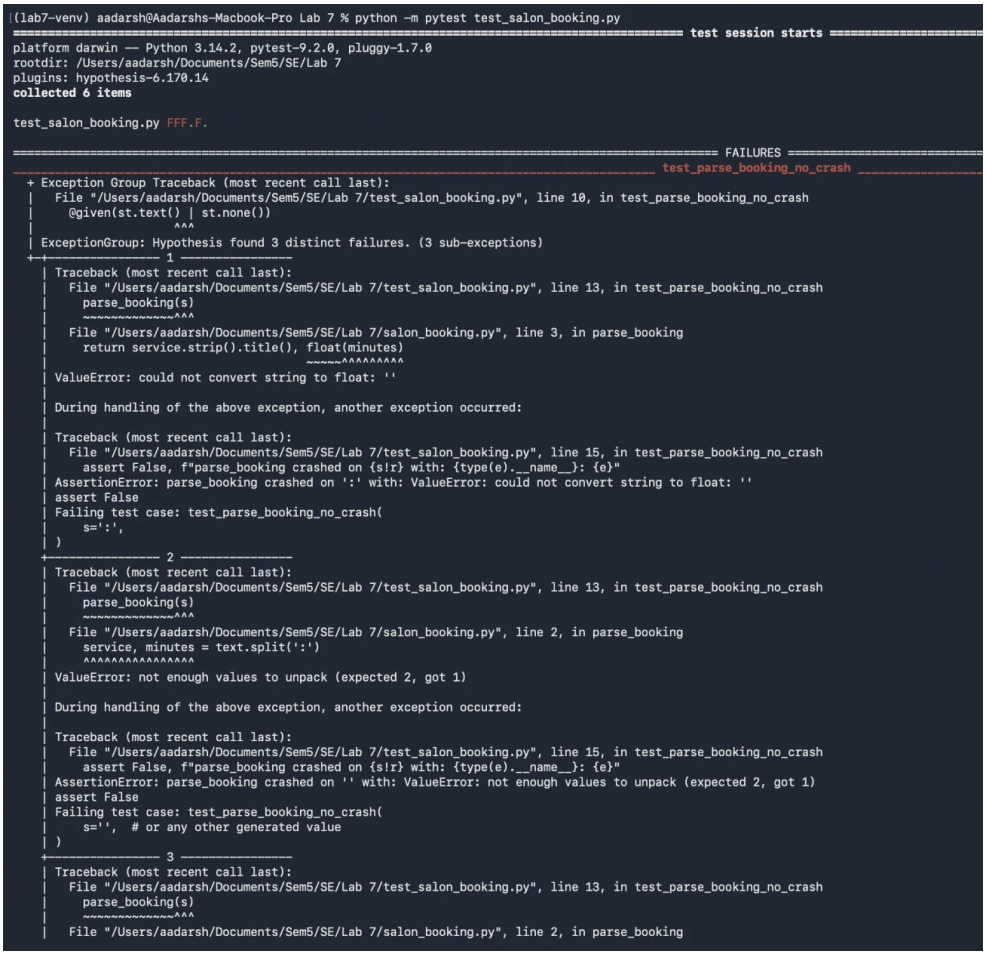
Command: `python salon_booking.py`. Entering `haircut` (no colon) crashes `parse_booking` with a ValueError.



Step 3 & 4: Hypothesis tests run against the buggy code

Command: `python -m pytest test_salon_booking.py`. Result: **4 failed, 2 passed**. Hypothesis found these failing inputs:

- **parse_booking**: crashes on `':'`, `''` and `None`
- **duration_average**: crashes on `''` and `None`
- **payable_bill**: crashes on `'|'`, `''` and `None`
- **duration_average**: wrong result, average of [1, 2] returned 1 instead of 1.5 (integer floor division)



```
(lab7-venv) aadarsh@Aadarshs-Macbook-Pro Lab 7 % python -m pytest test_salon_booking.py
============================= test session starts ==============================
platform darwin -- Python 3.14.2, pytest-9.2.0, pluggy-1.7.0
rootdir: /Users/aadarsh/Documents/Sem5/SE/Lab 7
plugins: hypothesis-6.170.14
collected 6 items

test_salon_booking.py FFF.F.

=================================== FAILURES ===================================
_____________________ test_parse_booking_no_crash ______________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/aadarsh/Documents/Sem5/SE/Lab 7/test_salon_booking.py", line 10, in test_parse_booking_no_crash
  |     @given(st.text() | st.none())
  |                        ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/aadarsh/Documents/Sem5/SE/Lab 7/test_salon_booking.py", line 13, in test_parse_booking_no_crash
    |     parse_booking(s)
    |     ~~~~~~~~~~~~~^^^
    |   File "/Users/aadarsh/Documents/Sem5/SE/Lab 7/salon_booking.py", line 3, in parse_booking
    |     return service.strip().title(), float(minutes)
    |                                     ~~~~~^^^^^^^^^
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/aadarsh/Documents/Sem5/SE/Lab 7/test_salon_booking.py", line 15, in test_parse_booking_no_crash
    |     assert False, f"parse_booking crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: parse_booking crashed on ':' with: ValueError: could not convert string to float: ''
    | assert False
    | Failing test case: test_parse_booking_no_crash(
    |     s=':',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/aadarsh/Documents/Sem5/SE/Lab 7/test_salon_booking.py", line 13, in test_parse_booking_no_crash
    |     parse_booking(s)
    |     ~~~~~~~~~~~~~^^^
    |   File "/Users/aadarsh/Documents/Sem5/SE/Lab 7/salon_booking.py", line 2, in parse_booking
    |     service, minutes = text.split(':')
    |     ^^^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/aadarsh/Documents/Sem5/SE/Lab 7/test_salon_booking.py", line 15, in test_parse_booking_no_crash
    |     assert False, f"parse_booking crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: parse_booking crashed on '' with: ValueError: not enough values to unpack (expected 2, got 1)
    | assert False
    | Failing test case: test_parse_booking_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "/Users/aadarsh/Documents/Sem5/SE/Lab 7/test_salon_booking.py", line 13, in test_parse_booking_no_crash
    |     parse_booking(s)
    |     ~~~~~~~~~~~~~^^^
    |   File "/Users/aadarsh/Documents/Sem5/SE/Lab 7/salon_booking.py", line 2, in parse_booking
```

```
    |     service, minutes = text.split(':')
    |                        ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/aadarsh/Documents/Sem5/SE/Lab 7/test_salon_booking.py", line 15, in test_parse_booking_no_crash
    |     assert False, f"parse_booking crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: parse_booking crashed on None with: AttributeError: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_parse_booking_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_____ test_duration_average_no_crash _____
  + Exception Group Traceback (most recent call last):
  |   File "/Users/aadarsh/Documents/Sem5/SE/Lab 7/test_salon_booking.py", line 19, in test_duration_average_no_crash
  |     @given(st.text() | st.none())
  |                ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/aadarsh/Documents/Sem5/SE/Lab 7/test_salon_booking.py", line 22, in test_duration_average_no_crash
    |     duration_average(s)
    |     ~~~~~~~~~~~~~~~~^^^
    |   File "/Users/aadarsh/Documents/Sem5/SE/Lab 7/salon_booking.py", line 6, in duration_average
    |     values = [int(r) for r in durations_csv.split(',')]
    |               ~~~^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/aadarsh/Documents/Sem5/SE/Lab 7/test_salon_booking.py", line 24, in test_duration_average_no_crash
    |     assert False, f"duration_average crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: duration_average crashed on '' with: ValueError: invalid literal for int() with base 10: ''
    | assert False
    | Failing test case: test_duration_average_no_crash(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         /Users/aadarsh/Documents/Sem5/SE/Lab 7/test_salon_booking.py:23
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/aadarsh/Documents/Sem5/SE/Lab 7/test_salon_booking.py", line 22, in test_duration_average_no_crash
    |     duration_average(s)
    |     ~~~~~~~~~~~~~~~~^^^
    |   File "/Users/aadarsh/Documents/Sem5/SE/Lab 7/salon_booking.py", line 6, in duration_average
    |     values = [int(r) for r in durations_csv.split(',')]
    |                              ^^^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/aadarsh/Documents/Sem5/SE/Lab 7/test_salon_booking.py", line 24, in test_duration_average_no_crash
    |     assert False, f"duration_average crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: duration_average crashed on None with: AttributeError: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_duration_average_no_crash(
```

```
    |     s=None,  # or any other generated value
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         /Users/aadarsh/Documents/Sem5/SE/Lab 7/test_salon_booking.py:23
    +------------------------------------
_____ test_payable_bill_no_crash _____
  + Exception Group Traceback (most recent call last):
  |   File "/Users/aadarsh/Documents/Sem5/SE/Lab 7/test_salon_booking.py", line 28, in test_payable_bill_no_crash
  |     @given(st.text() | st.none())
  |                ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/aadarsh/Documents/Sem5/SE/Lab 7/test_salon_booking.py", line 31, in test_payable_bill_no_crash
    |     payable_bill(s)
    |     ~~~~~~~~~~~~^^^
    |   File "/Users/aadarsh/Documents/Sem5/SE/Lab 7/salon_booking.py", line 11, in payable_bill
    |     total = float(bill) * 1.18 - float(discount)
    |             ~~~~~^^^^^^
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/aadarsh/Documents/Sem5/SE/Lab 7/test_salon_booking.py", line 33, in test_payable_bill_no_crash
    |     assert False, f"payable_bill crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: payable_bill crashed on '|' with: ValueError: could not convert string to float: ''
    | assert False
    | Failing test case: test_payable_bill_no_crash(
    |     s='|',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/aadarsh/Documents/Sem5/SE/Lab 7/test_salon_booking.py", line 31, in test_payable_bill_no_crash
    |     payable_bill(s)
    |     ~~~~~~~~~~~~^^^
    |   File "/Users/aadarsh/Documents/Sem5/SE/Lab 7/salon_booking.py", line 10, in payable_bill
    |     bill, discount = text.split('|')
    |     ^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/aadarsh/Documents/Sem5/SE/Lab 7/test_salon_booking.py", line 33, in test_payable_bill_no_crash
    |     assert False, f"payable_bill crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: payable_bill crashed on '' with: ValueError: not enough values to unpack (expected 2, got 1)
    | assert False
    | Failing test case: test_payable_bill_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "/Users/aadarsh/Documents/Sem5/SE/Lab 7/test_salon_booking.py", line 31, in test_payable_bill_no_crash
    |     payable_bill(s)
    |     ~~~~~~~~~~~~^^^
    |   File "/Users/aadarsh/Documents/Sem5/SE/Lab 7/salon_booking.py", line 10, in payable_bill
    |     bill, discount = text.split('|')
    |                      ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
```

```
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/aadarsh/Documents/Sem5/SE/Lab 7/test_salon_booking.py", line 33, in test_payable_bill_no_crash
    |     assert False, f"payable_bill crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: payable_bill crashed on None with: AttributeError: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_payable_bill_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_____________________________________________________________________ test_duration_average_valid_input _____________________

    @given(st.lists(st.integers(min_value=1, max_value=600), min_size=1, max_size=20))
>   def test_duration_average_valid_input(durations):
                ^^^

test_salon_booking.py:52:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

durations = [1, 2]

    @given(st.lists(st.integers(min_value=1, max_value=600), min_size=1, max_size=20))
    def test_duration_average_valid_input(durations):
        result = duration_average(",".join(map(str, durations)))
        # the average must lie between the smallest and largest duration
        assert min(durations) <= result <= max(durations)
>       assert math.isclose(result, sum(durations) / len(durations), abs_tol=0.01)
E       assert False
E        +  where False = <built-in function isclose>(1, (3 / 2), abs_tol=0.01)
E        +    where <built-in function isclose> = math.isclose
E        +    and   3 = sum([1, 2])
E        +    and   2 = len([1, 2])
E       Failing test case: test_duration_average_valid_input(
E           durations=[1, 2],
E       )

test_salon_booking.py:56: AssertionError
=========================================================================================== short test summary info ============
FAILED test_salon_booking.py::test_parse_booking_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_salon_booking.py::test_duration_average_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_salon_booking.py::test_payable_bill_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_salon_booking.py::test_duration_average_valid_input - assert False
========================================================================================== 4 failed, 2 passed in 1.57s =========
(lab7-venv) aadarsh@Aadarshs-Macbook-Pro Lab 7 %
```

Step 5: Run the fixed application

Each function now validates its input and returns None instead of crashing; main() prints a clear error message. Invalid inputs (`haircut`, `30,,abc`, `nan|5`):

```
(lab7-venv) aadarsh@Aadarshs-Macbook-Pro Lab 7 % cp salon_booking_fixed.py salon_booking.py
(lab7-venv) aadarsh@Aadarshs-Macbook-Pro Lab 7 % python salon_booking.py
==== Running parse_booking ====
Enter service:duration in min (e.g. haircut:45): haircut
Invalid input. Use the format service:minutes, e.g. haircut:45

==== Running duration_average ====
Enter durations as CSV (e.g. 30,45,60): 30,,abc
Invalid input. Use positive whole numbers separated by commas, e.g. 30,45,60

==== Running payable_bill ====
Enter bill|discount (e.g. 800|100): nan|5
Invalid input. Use the format bill|discount with non-negative numbers, e.g. 800|100
(lab7-venv) aadarsh@Aadarshs-Macbook-Pro Lab 7 %
```

Valid inputs (`haircut:45`, `30,45,60`, `800|100`):

```
(lab7-venv) aadarsh@Aadarshs-Macbook-Pro Lab 7 % python salon_booking.py
==== Running parse_booking ====
Enter service:duration in min (e.g. haircut:45): haircut:45
Service and Duration: ('Haircut', 45.0)

==== Running duration_average ====
Enter durations as CSV (e.g. 30,45,60): 30,45,60
Average Duration: 45.0

==== Running payable_bill ====
Enter bill|discount (e.g. 800|100): 800|100
Payable Bill: 844.0
(lab7-venv) aadarsh@Aadarshs-Macbook-Pro Lab 7 %
```

Step 6: Re-run the tests against the fixed code

Command: `python -m pytest test_salon_booking.py -v`. Result: **6 passed**.

```
(lab7-venv) aadarsh@Aadarshs-Macbook-Pro Lab 7 % python -m pytest test_salon_booking.py -v
============================= test session starts ==============================
platform darwin -- Python 3.14.2, pytest-9.2.0, pluggy-1.7.0 -- /Users/aadarsh/lab7-venv/bin/python
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: /Users/aadarsh/Documents/Sem5/SE/Lab 7
plugins: hypothesis-6.170.14
collected 6 items

test_salon_booking.py::test_parse_booking_no_crash PASSED
test_salon_booking.py::test_duration_average_no_crash PASSED
test_salon_booking.py::test_payable_bill_no_crash PASSED
test_salon_booking.py::test_parse_booking_valid_input PASSED
test_salon_booking.py::test_duration_average_valid_input PASSED
test_salon_booking.py::test_payable_bill_valid_input PASSED

============================== 6 passed in 0.36s ===============================
(lab7-venv) aadarsh@Aadarshs-Macbook-Pro Lab 7 %
```